1. a) Write down the formal definition of $\Theta(g(n))$. [10%]

b) Simplify the following functions by expressing them in Θ-notation. Give a formal justification for each case, referring to the definition of Θ-notation.

(i) $2n^3 - 3n^2 + 5n$

(ii) $2n \log n - 5 \log n$

[30%]

c) Copy the following table to your answer booklet and indicate for each pair of functions $f(n), g(n)$ whether $f(n)$ is O, o, Ω, ω, or Θ of $g(n)$ by marking **all** cells that apply.

$f(n)$	$g(n)$	O	o	Ω	ω	Θ
n	$10n$					
n^2	$\sqrt{n}$					
$\log n$	$n \log n$					
1000	$\log n$					
$n \sum_{i=1}^{n} i$	n^2					
$(\log n)^2$	$n^{0.1}$					

[30%]

d) The following algorithm makes sure that all numbers in an array $A[1 \ldots n]$ of length n have a value of at least 9.50.

```
MINIMUM-WAGE-CORRECTOR(A)
for i = 1 to A.length do
    if A[i] < 9.50 then
        A[i] = 9.50
```

Prove the correctness of MINIMUM-WAGE-CORRECTOR by stating an appropriate loop invariant and showing the three properties: initialisation, maintenance, and termination.

[30%]

2. a) Consider the execution of COUNTINGSORT on the following input array A. Recall that during the final for loop the input elements are written to an output array B based on counters stored in an array C whose contents just before the final loop are given below.

A	1	3	2	1	0	3	1

C	1	4	5	7

Copy the table below to your answer booklet and write down the contents of the arrays B and C at the end of each iteration of the final for loop.

Array C				Array B						
0	**1**	**2**	**3**	**1**	**2**	**3**	**4**	**5**	**6**	**7**

[20%]

b) Given an array A of size n containing numbers, design an algorithm that outputs the largest $m(< n)$ numbers of the array.

(i) First design an algorithm that has time complexity $O(n \log n)$. [10%]

(ii) Improve the time complexity of your algorithm to $O(n \log m)$ by using the *Heap* data structure. You are required to design an algorithm with time complexity $O(n \log m)$ (either in pseudocode or in English) and analyse the correctness of it using a loop invariant. [30%]

c) A list of n different sized pegs and a list of n different sized holes are given to us with the guarantee that each peg fits in a hole perfectly. We have to match each peg to a hole that fits it perfectly.

The *caveat* is the following: We cannot compare two pegs to find out which one is bigger. Nor can we do it for two holes. The only comparisons allowed are comparisons between pairs consisting of a peg and a hole to find out whether the hole is bigger or smaller for the peg.

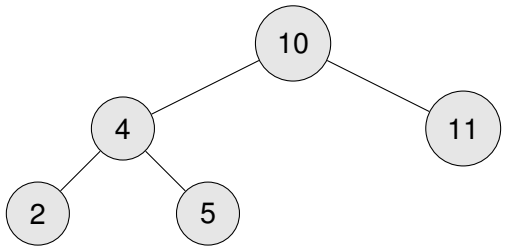
Design an algorithm (by modifying RANDOMIZED-QUICKSORT) that solves this problem in time complexity $O(n \log n)$. [40%]

3. a) Implement the *Queue* data structure using *Stack* data structure. You can use two stacks. In addition, you must implement at least one of the DEQUEUE and ENQUEUE operations with time complexity $O(1)$.

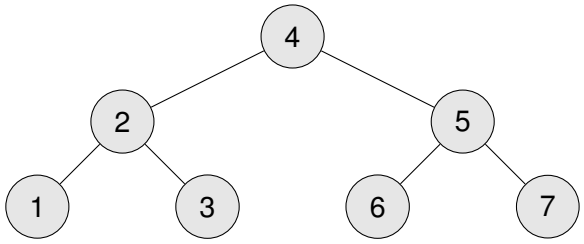
   (i) Present an algorithm that performs the DEQUEUE and ENQUEUE operations (either in English or in pseudocode). [20%]

   (ii) What are the time complexity of these two operations? [10%]

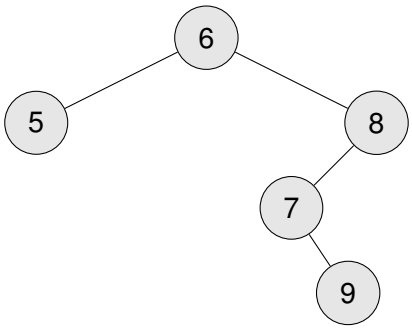
   b) For each of the following trees, explain which show binary search trees.



(i) [10%]



(ii) [10%]



(iii) [10%]

   c) Given two nodes u and v in a rooted tree, the *lowest common ancestor* or LCA of u and v is the lowest node w (farthest away from the root) that has both u and v as descendants. For example, in the figure associated with Question 3(b)(i), the LCA of 2 and 5 is 4, and the LCA of 5 and 11 is 10.

   Design an algorithm (in pseudocode or in English) by modifying the TREE-SEARCH algorithm that finds the LCA of two nodes u and v (given as input) in a binary search tree T. If the height of T is h, your algorithm should have time complexity $O(h)$. Prove the correctness of your algorithm (You can assume the correctness of the TREE-SEARCH algorithm). [40%]

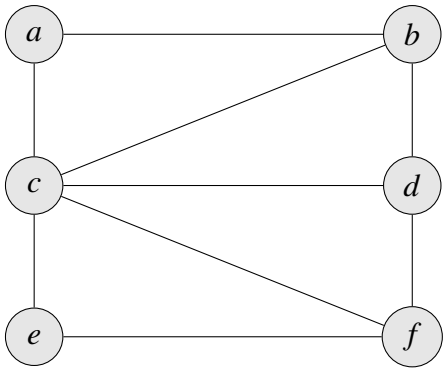
4. a) The following matrix shows the costs of fibre optic broadband connections between five cities.

$$\begin{pmatrix} \infty & 2 & \infty & \infty & 7 \\ 2 & \infty & 4 & 3 & 8 \\ \infty & 4 & \infty & 5 & 9 \\ \infty & 3 & 5 & \infty & 10 \\ 7 & 8 & 9 & 10 & \infty \end{pmatrix}$$

(i) Draw a diagram of this graph (omit edges of ∞ cost). [10%]

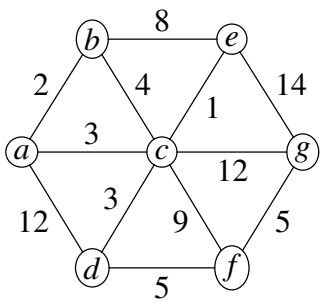
(ii) Find the cheapest network connecting all cities and give its connection cost. State which algorithm you used. [20%]

b) Apply Breadth-first search starting at node a in the graph below. Assume that all adjacency lists are sorted alphabetically. Show the resulting tree and print the contents of the queue at each step that a node becomes black.



[30%]

c) Use Dijkstra's algorithm to work out the shortest path from a to g in the graph below. Show the working of the algorithm.



[30%]

d) Explain (briefly) why Prim's algorithm is a greedy algorithm. [10%]

END OF QUESTION PAPER